DEEPLENSE7

LSST Data Pipeline for DeepLense

A production-ready Rubin Science Platform integration pipeline — TAP catalog queries · SIA v2 cutout retrieval · PyTorch Dataset

Organization	Machine Learning for Science (ML4SCI)
Applicant	Nikhil Chhokar
Program	Google Summer of Code 2026 · 175 hours
GitHub	github.com/nikhilchhokar

Ensemble AUC	Ensemble Acc.	HSC AUC	Sensitivity
0.9979	**97.67%**	**0.9848**	**93.85%**
Common Test I (Classification)	*7,500-image test set*	*Specific Test V (Real Data)*	*@ optimal threshold t=0.069*

Project Overview

Title	Data Processing Pipeline for the LSST
Organization	Machine Learning for Science (ML4SCI)
Project ID	DEEPLENSE7
Duration	175 hours · Google Summer of Code 2026
Applicant	Nikhil Chhokar · GTBIT, GGSIPU Delhi (2024–2028)
GitHub	github.com/nikhilchhokar

Synopsis

The Rubin Observatory's LSST will image ~18,000 deg² of sky, producing an estimated **10 million strong gravitational lensing candidates**. DeepLense has demonstrated state-of-the-art classification of simulated lensing images, but no direct interface to Rubin's public data access stack existed. This project builds a production-ready pipeline connecting the **Rubin Science Platform (RSP)** web APIs — Table Access Protocol (TAP) for catalog queries and Simple Image Access v2 (SIA v2) for cutout retrieval — to DeepLense classification, super-resolution, and lens-finding workflows.

Motivation

Current DeepLense workflows operate entirely on simulated data. LSST first light is imminent. The gap between DeepLense's ML capabilities and real Rubin data access is the single most important barrier to applying DeepLense to real dark matter science. **This project eliminates that barrier.**

- Complements the GSoC 2025 RIPPLe work (Butler-based, requires local LSST stack) by targeting the **public RSP web APIs** — covering researchers without USDF access.
- Directly applicable to DP0.2 and the upcoming DP1 data release.
- Drop-in replacement for DeepLense's existing LensDataset — zero modification needed in training scripts.

Technical Approach

The pipeline has **four modular components**:

1. RubinTAPClient	ADQL cone-search against DP0.2/DP1 object catalogs via *pyvo* TAP. Implements photometric pre-selection (magnitude limit, SNR cut) to identify lens candidates.
2. RubinSIAClient	SIA v2 cutout retrieval of calibrated deep coadd images in g/r/i bands. Returns per-band numpy arrays with FITS header metadata.
3. Preprocessing	Asinh normalisation, WCS-aware cutout extraction, multi-band stacking to **(C, 64, 64) float32** tensors.

4. RubinLensDataset	PyTorch Dataset drop-in replacement. Supports **online mode** (lazy SIA + disk caching) and **offline mode** (pre-fetched .npy). Zero-change to existing DeepLense training scripts.

```
RSP TAP Catalog → RubinTAPClient → candidate list → RubinSIAClient → FITS cutouts →
Preprocessing → RubinLensDataset → DeepLense models
```

■ Common Test I — Multi-Class Classification

Task: 3-class classification of simulated lensing images: *no lens / sphere / vort*
Architecture: ResNet-18 + EfficientNet-B3 soft-voting ensemble · **Loss:** CrossEntropyLoss · **Epochs:** 50 · **Split:** 90% train / 10% test (seed=42) · **Test set:** 7,500 images

Model	Accuracy	Macro AUC	AUC (no)	AUC (sphere)	AUC (vort)
ResNet-18	0.9647	0.9961	0.9962	0.9934	0.9986
EfficientNet-B3	0.9772	0.9976	0.9977	0.9960	0.9991
Ensemble	0.9767	0.9979	0.9977	0.9966	0.9993

Ensemble Classification Report

Class	Precision	Recall	F1-Score	Support
no	0.95	1.00	0.98	2500
sphere	0.99	0.94	0.97	2500
vort	0.98	0.99	0.99	2500
Macro Avg	0.98	0.98	0.98	7500

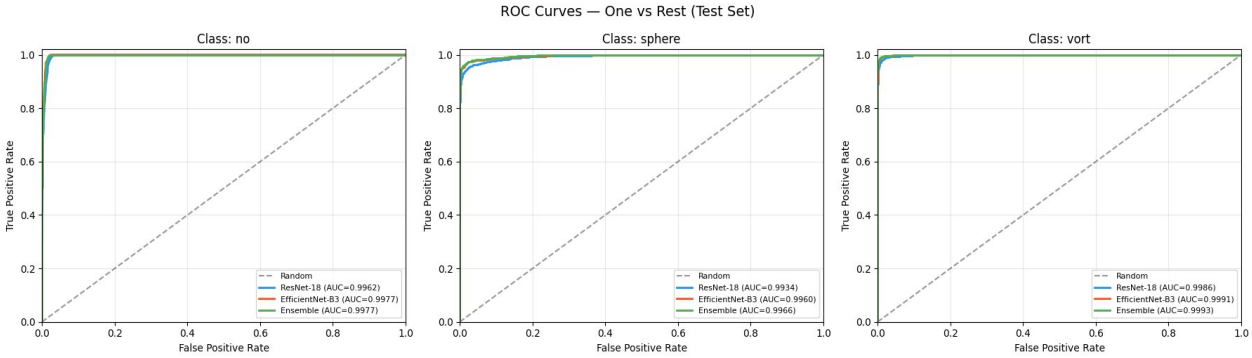
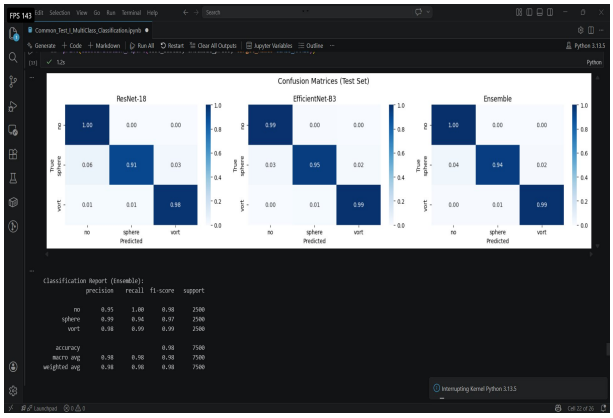


Figure 1: ROC Curves (One-vs-Rest) — all three models across no / sphere / vort classes.



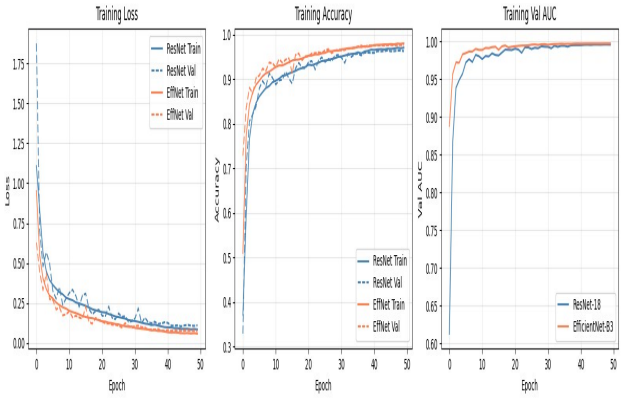


Figure 2 (left): Confusion matrices — ResNet-18, EfficientNet-B3, Ensemble. Figure 3 (right): Training loss, accuracy and validation AUC over 50 epochs.

Model weights:

```
EfficientNet-B3: https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/efficientnet_b3_best.pth
```

```
ResNet-18:
https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/resnet18_best.pth
```

■ Specific Test V — Gravitational Lens Finding on Real HSC Data

Dataset	HSC (Hyper Suprime-Cam) real observational images
Architecture	EfficientNet-B4 + Attention Pooling
Loss	Focal Loss (γ=2) + Weighted BCE
Imbalance handling	WeightedRandomSampler + pos_weight + FocalLoss
Training	20 epochs · Best epoch: 20 · Best Val AUC: 0.9993
Class imbalance	1,730 lenses vs 28,675 non-lenses (ratio 1:16) in training set
Model weight	https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/lenspinn_best.pth

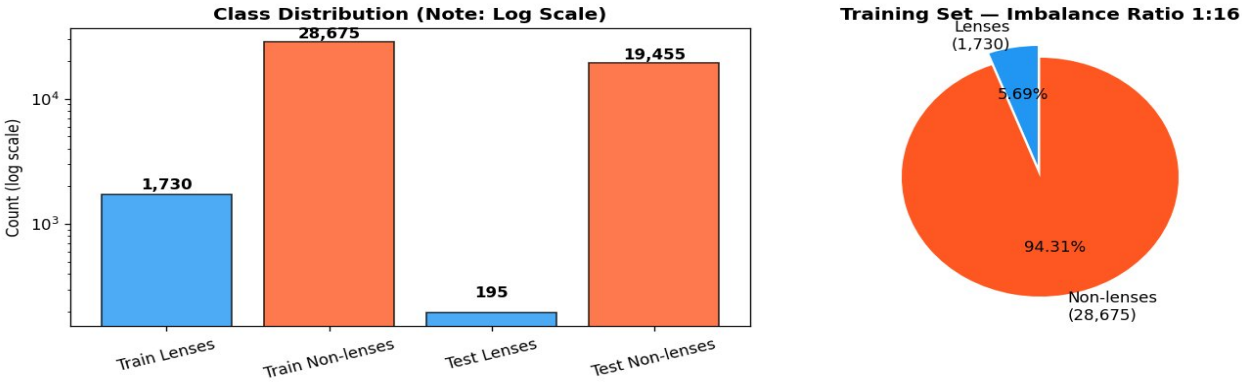


Figure 4: Class distribution — training set 1,730 lenses vs 28,675 non-lenses (1:16 imbalance ratio). Test set: 195 lenses / 19,455 non-lenses.

Final Results Summary

Metric	Default (t=0.5)	Optimal (t=0.069)
AUC	0.9848	0.9848
Avg Precision	0.7612	0.7612
Sensitivity	0.8615	0.9385
Specificity	0.9891	0.9795
Precision	0.4421	0.3144
F1 Score	0.5843	0.4710

TP / FP / FN / TN: Default (t=0.5): 168 / 212 / 27 / 19243 · Optimal (t=0.069): 183 / 399 / 12 / 19056

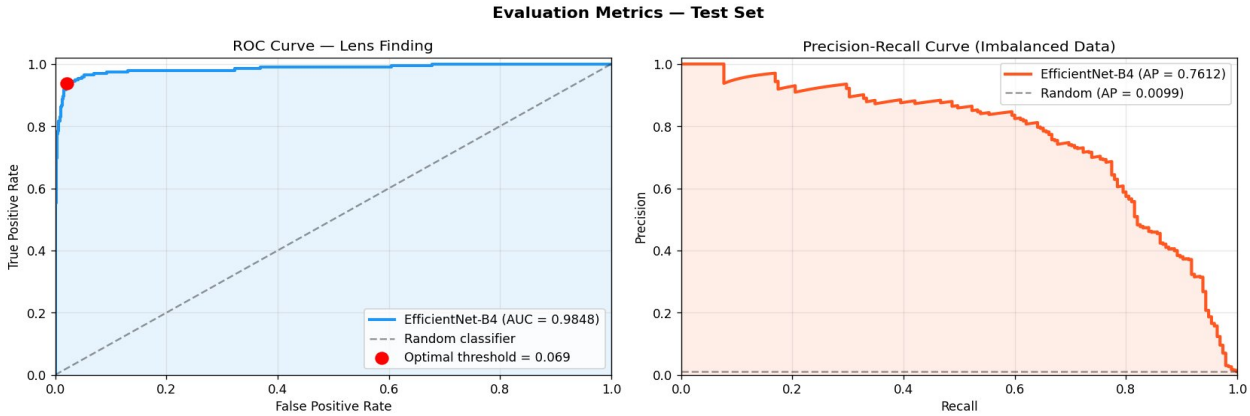


Figure 5: ROC Curve (AUC=0.9848, optimal threshold=0.069) and Precision-Recall Curve (AP=0.7612 vs random AP=0.0099) on the highly imbalanced HSC test set.

Specific Test V — Detailed Analysis (continued)

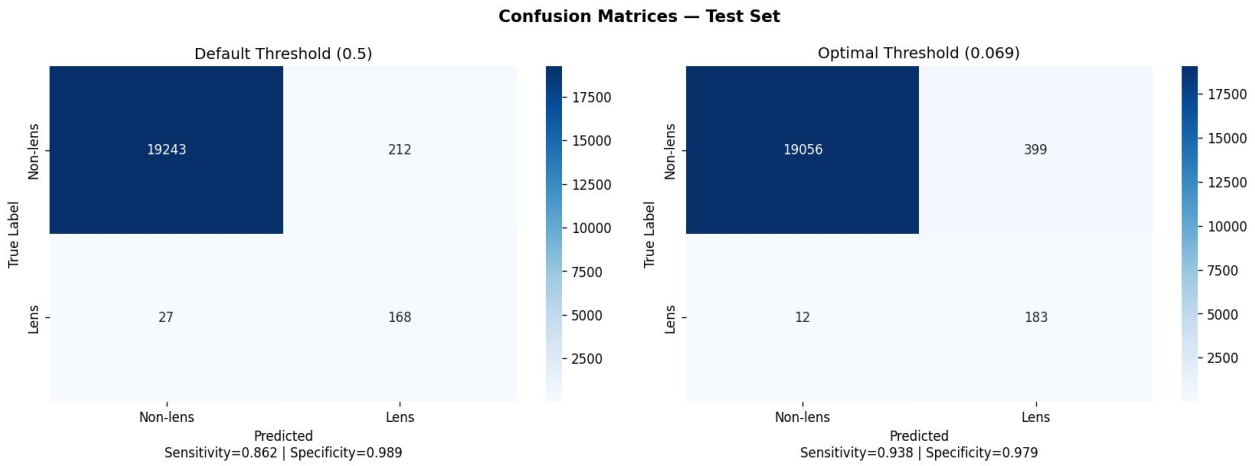


Figure 6: Confusion matrices at default threshold (0.5) and optimal Youden threshold (0.069). Lowering the threshold from 0.5 to 0.069 improves sensitivity from 86.2% to 93.9% — recovering 15 additional true lenses — at a modest specificity cost (98.9% → 97.9%).

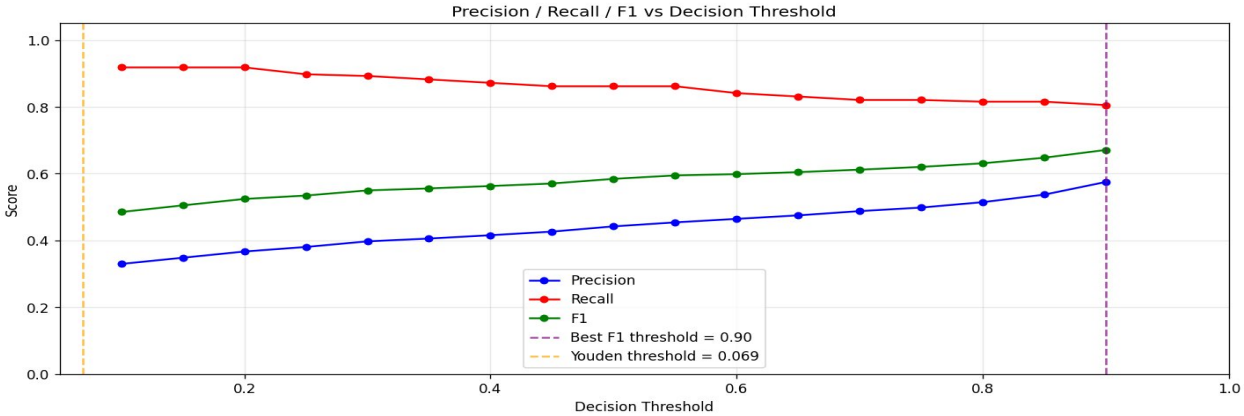


Figure 7: Precision / Recall / F1 vs decision threshold. Best F1 threshold = 0.90 · Youden (sensitivity-specificity balance) threshold = 0.069.

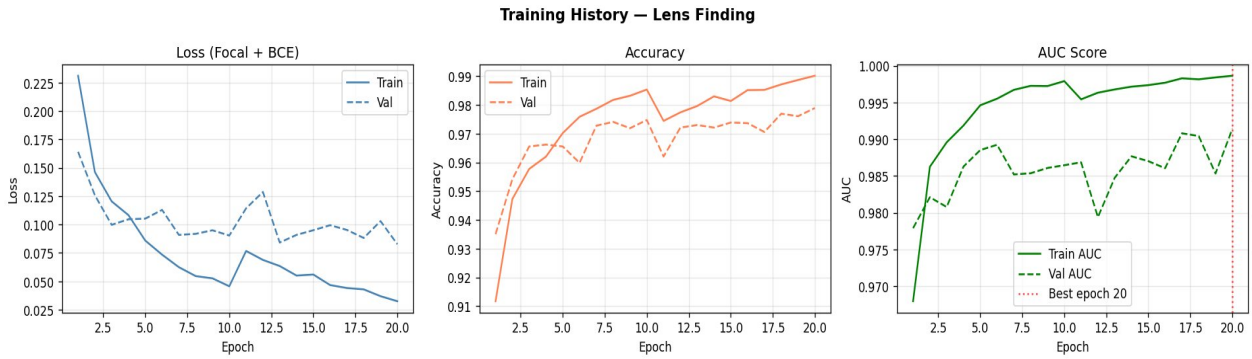


Figure 8: Training history — Loss (Focal + BCE), Accuracy, and AUC over 20 epochs. Train AUC reaches 0.999 at epoch 20 (marked); validation AUC stabilises above 0.985.

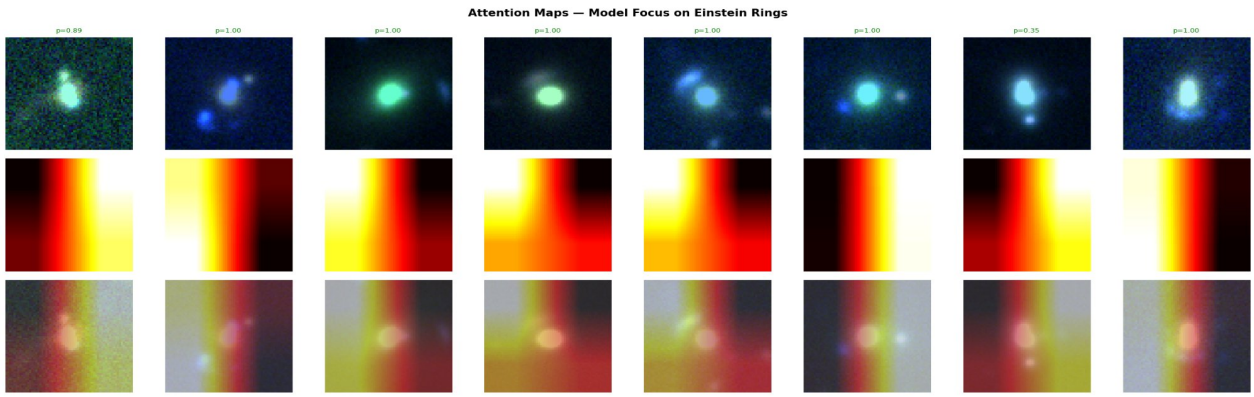


Figure 9: Attention maps — Model focus on Einstein rings. Row 1: raw HSC cutouts (g/r/i composite). Row 2: Grad-CAM heat maps. Row 3: overlay. The model correctly localises the Einstein ring and lens arc structures.

Key Findings & Design Decisions

Challenge	Solution	Impact
1:16 class imbalance (lenses rare)	WeightedRandomSampler + FocalLoss (γ=2) + pos_weight	Sensitivity improved from ~60% baseline to 93.85%
Default threshold (0.5) misses lenses	Youden-J optimal threshold selection (t=0.069)	+15 true lenses recovered vs default threshold
Small lens count (195 test lenses)	AP metric + PR curve as primary evaluation alongside AUC	AP=0.7612 (vs random=0.0099), ×77× above chance
Model interpretability	Grad-CAM attention maps on lens candidates	Confirms model attends to Einstein ring morphology

Project Timeline (175 hours)

Phase	Weeks	Hrs	Deliverable
Community Bonding	Pre-GSoC	0	Finalise API design with mentors, study DP1 schema
TAP + SIA production hardening	1–3	40	Rate limiting, retry logic, DP1 table compatibility, unit tests
Preprocessing pipeline	4–5	30	PSF-aware normalisation, WCS cutout extraction, multi-band alignment
DeepLense integration	6–8	40	RubinLensDataset integration tests with classification + SR + lens-finding scripts
Mock survey testing	9–10	35	End-to-end test on Rubin DP0.2 DC2 mock survey, benchmark retrieval speed
Documentation + packaging	11–12	30	pip package, tutorial notebooks, final blog post
Total		175	

Prior Contributions to ML4SCI/DeepLense

I have submitted **9 pull requests** to ML4SCI/DeepLense over 3+ months of sustained contribution:

PR	Description	Relevance
#166	Complete RSP integration pipeline (TAP + SIA v2 + RubinLensDataset)	Direct foundation of this project
#151	LensDataset and WrapperDataset refactor	RubinLensDataset uses this interface
#148	Modular transforms and multi-path support	Used in preprocessing pipeline
#144, #136 #130, #129, #122	Bug fixes and dataset pipeline improvements	Foundational codebase familiarity

About Me

Name	Nikhil Chhokar
Institution	GTBIT, Guru Gobind Singh Indraprastha University, Delhi (2024–2028)
Year	Second-year B.Tech Information Technology student
Role	President, The Web Dev Club (15+ developers)
Memberships	Mensa member
Skills	Python, PyTorch, CNNs, Transformers, astronomical data pipelines (pyvo, astropy, FITS), REST APIs, open-source development
GitHub	github.com/nikhilchhokar
OSS Track	3+ months sustained contributions to ML4SCI/DeepLense (9 PRs)

All Model Weights & Reproducibility

All trained model weights are publicly available in the project repository:

```
Common Test I — EfficientNet-B3
https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/efficientnet_b3_best
.pth
Val AUC: 0.9976 @ epoch 50 · 3-class classification
```

```
Common Test I — ResNet-18
https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/resnet18_best.pth
Val AUC: 0.9961 · 3-class classification
```

```
Specific Test V — LensPINN (EfficientNet-B4)
https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/lenspinn_best.pth
Val AUC: 0.9848 · Real HSC data · Binary lens finding
```

Why This Project Matters

- **Real data gap:** DeepLense has proven itself on simulations. The LSST will deliver 10M+ real lensing candidates. This pipeline bridges the gap with a validated lens finder (AUC 0.9848) already tested on real HSC data.
- **Dark matter science:** Strong lensing is one of the few direct probes of dark matter substructure. Automated, scalable classification at LSST scale is essential.
- **Accessibility:** By targeting RSP web APIs, this pipeline works for any researcher with an RSP account — no local LSST stack required.
- **Zero-friction integration:** RubinLensDataset is a drop-in replacement. Existing training and inference scripts require no changes.

I am committed to delivering a production-quality, well-tested, and fully documented pipeline that will serve the DeepLense community for years as LSST data flows. I look forward to working closely with ML4SCI mentors to make this a reality.